

Seyed Kayvan Abbasi

CONTACT INFORMATION

121 Wooded Lane,
Villanova, PA 19085

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EDUCATION

M.Sc. Mechanical Engineering, Villanova University
Thesis title: Analysis of external system components for
thermodynamic modeling of data centers
Full Time Graduate Research Assistant
GPA (until present): 4.0/4.0
Advisor: Dr. Aaron P. Wemhoff, Co-Advisor: Dr. Alfonso Ortega

Jan. 2011-
present

M.Sc. Mechanical Engineering, Amirkabir University(www.aut.ac.ir)
Thesis title: Generating a homogeneous isotropic turbulent box using a
three dimensional turbulent solver in Fourier space (LES approach)
GPA: 15.13/20
Advisor: Dr. Bamdad Lessani

2005-2008

B.Sc. Mechanical Engineering, K.N.T.U University (www.kntu.ac.ir)
Thesis title: Developing an object-oriented code to evaluate
conductance and evacuation time for a vacuum chamber with its joints
and pipes
GPA: 14.39/20
Advisor: Dr. Mehrzad Shams

2000-2005

RESEARCH INTERESTS

Thermodynamic system modeling, Nano scale heat transfer using Molecular Dynamics (MD), Computational Fluid Dynamics, Simulation of Turbulent Flows (DNS, LES), Turbulent Combustion Modeling, Two phase flows (Especially Gas-Solid flow), Compressible flows (Also Low-Mach Number flows)

HONORS

- Ranked 1st among 20,000 candidates of Master of Science entrance exam for Azad University in Iran
- Ranked in top 1% (with the rank 2604 out of 313440 candidates) of nation-wide entrance exam for Bachelor of Science in Iranian universities

PUBLICATIONS

Poster Presentations

- **Kayvan Abbasi**, Marcelo Del Valle, Aaron P. Wemhoff and Alfonso Ortega, "Thermodynamic Modeling of Data Center Cooling Systems Part 2: Component Models", NSF I/UCRC meeting poster session, University of Texas at Arlington, TX, October 2013
- **Kayvan Abbasi**, Aaron P. Wemhoff and Alfonso Ortega, "First Generation Chiller And Cooling Tower Models For Exergy Analysis Modeling", NSF I/UCRC meeting poster session, Villanova Conference Center, Villanova, June 2013

Conference Papers under Preparation

- **Abbasi, K.**, Wemhoff, A.P., and Ortega, A., "A Transient Model for Heat Exchangers Containing Moist Air", Proceedings of the 2013 ASME InterPACK Conference, (IPACK2013), San Francisco, CA, July 16-18, 2013.
- **Abbasi, K.**, Wemhoff, A.P., and Ortega, A., "A Detailed Model Of Steady State Counterflow Mechanical Draft Cooling Towers", Proceedings of the 2013 ASME InterPACK Conference, (IPACK2013), San Francisco, CA, July 16-18, 2013.
- Wemhoff, A.P., Marcelo Del Valle, **Abbasi, K.**, , and Ortega, A., "Thermodynamic Modeling Of Data Center Cooling Systems", Proceedings of the 2013 ASME InterPACK Conference, (IPACK2013), San Francisco, CA, July 16-18, 2013.

Author/Co-author of 2 journal papers under Preparation

- **Abbasi, K.**, Wemhoff, A.P., and Ortega, A., "A Detailed Model Of Steady State Counterflow Mechanical Draft Cooling Towers"
- Wemhoff, A.P., Marcelo Del Valle, **Abbasi, K.**, , and Ortega, A., "Thermodynamic Modeling Of Data Center Cooling Systems"

PROFESSIONAL ACTIVITIES	• Technical Reviewer of Journal of Nanoscale and Microscale Thermophysical Engineering and ASME Conferences • ASME Student Member	
WORKING EXPERIENCE	• Taban Chekad Company -Iran - summer of 2005-Worked as consulting engineer in designing a Rolling Machine Stand • Alkan Machine Company- Iran- Summer of 2006-Worked as intern engineer in designing a Gearbox for Cooling Tower	Feb. 2005-Aug.2005 Feb. 2006-Aug. 2006
SKILLS	Operating Systems: Windows, Linux Software Packages: Fluent, EES, Tecplot, Mathematica, AutoCAD, Office Suite CATIA (Degree from MFT), REFPROP Programming skills: MATLAB, FORTRAN, C++, Delphi, Visual Basic	
LANGUAGES	English: Fluent, TOEFL IBT Score :98, Reading: 22, Listening: 23, Speaking: 29, Writing: 24 GRE: Verbal: 147 Quantitative: 163 Farsi: Native Language, Arabic: Modest user	
EXTRACURRICULAR INTERESTS	Swimming; Playing Guitar	
REFERENCES	Dr. Aaron Wemhoff Assistant Professor, Department of Mechanical Engineering Villanova University College of Engineering (610) 519-8045 aaron.wemhoff@villanova.edu - Advisor for Master's Thesis at Villanova University Dr. Alfonso Ortega Associate Vice President for Research and Graduate Programs James R. Birle Professor of Energy Technology Villanova University College of Engineering (610) 519-7440 alfonso.ortega@villanova.edu - Co-Advisor for Master's Thesis at Villanova University Dr. Gerard Jones Professor and Associate Dean for Graduate Studies and Research (610) 519-4985 gerard.jones@villanova.edu - Instructor for Advanced Convection course Dr. Brian Sullivan Adjunct Professor, Department of Mechanical Engineering Villanova University College of Engineering brian.sullivan@villanova.edu - Instructor for Advanced Math II course Dr. Sridhar Santhanam Professor, Department of Mechanical Engineering Villanova University College of Engineering (610) 519-7924 sridhar.santhanam@villanova.edu - Instructor for Advanced Math course Dr. Bamdad Lessani Assistant Professor, Department of Mechanical Engineering Amirkabir University of Technology lessani@aut.ac.ir - Advisor for Master's degree at Amirkabir University of Technology	